

TONI ISMAIL

Software Engineer — Backend & Identity and Access Management (IAM)

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PROFESSIONAL SUMMARY

Software Engineer with nearly 4 years of experience building production-scale enterprise backends for Indonesia's largest fertilizer manufacturer (PT Pupuk Sriwidjaja Palembang), including the last 2 years leading a development team as Development Team Lead. Specialized in **Identity & Access Management**: built and operate a Keycloak-based Single Sign-On platform (OAuth 2.0, OpenID Connect, JWT, RBAC, WebAuthn/Passkeys) that serves as the single authentication gateway for 50+ internal applications and 3,000+ users. Proven record delivering large-scale REST APIs in **Golang (GoFiber)**, **NestJS**, and **Laravel** — including a procurement service with 338 endpoints and 54 domain models — backed by Kong API Gateway, Docker, and GitLab CI/CD pipelines with SonarQube quality gates. Lead 6 engineers in an Agile/Scrum practice, setting layered-architecture standards and mandatory OpenAPI documentation across all services.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, PHP, Python, SQL

Backend & Frameworks: GoFiber, NestJS, Laravel, Next.js (API Routes), Express

Identity & Access Management: Keycloak (gocloak Admin API), OAuth 2.0, OpenID Connect, JWT, WebAuthn / Passkeys, RBAC, SSO, single logout, OTP (WhatsApp)

APIs & Integration: REST, OpenAPI / Swagger, Kong API Gateway, SAP, webhooks, Google Sheets API, service-to-service HTTP clients

Databases & ORMs: PostgreSQL, SQL Server, MySQL, Redis, GORM, Prisma, Eloquent, schema migration & seeding

DevOps & Infrastructure: Docker, Docker Compose, GitLab CI/CD, SonarQube, Nexus Registry, Nginx, MinIO / S3-compatible storage, Cloudflare R2, Vercel

Testing & Quality: Jest (unit & e2e), Supertest, Go testing, smoke-test scripting, Postman, code review, static analysis

Architecture & Practices: Layered architecture (handler-service-repository), modular monolith, microservices, RBAC design, audit logging, caching strategy, security review & remediation, Agile/Scrum, technical documentation

Frontend (working knowledge): React.js, Next.js, React Native (Expo), Tailwind CSS

Tools: Git, GitLab, Postman, Jira, Figma, Swagger UI, Prisma Studio

PROFESSIONAL EXPERIENCE

PT Pupuk Sriwidjaja Palembang (PUSRI) — Palembang, Indonesia

Development Team Lead | August 2024 – Present

- Lead a team of 6 engineers across the full SDLC — sprint planning, code review, and production release — for four enterprise applications: **SIHPP**, **SIIRU**, **REKON**, and **SI DENDA**.
- Established cross-team engineering standards: layered handler-service-repository architecture, a consistent JSON response envelope, and mandatory OpenAPI documentation per service — cutting new-developer onboarding time and eliminating API-contract ambiguity with frontend teams.
- Designed a 4-stage GitLab CI/CD pipeline (build image → SonarQube static analysis → publish to Nexus registry → trigger infrastructure pipeline) across 3 branch environments, replacing manual releases with automated deployments where every merge must pass a quality gate.
- Architected and led two greenfield systems from zero to production: **SIHPP** (Golang/Fiber, SQL Server) and **SIIRU** (NestJS/Prisma, PostgreSQL, 68 endpoints) — both went live within 4–5 months.
- Translated requirements from Production Planning, Finance, and Logistics units into executable technical specifications; led requirement-gathering and UAT sessions with key users.
- Managed resource allocation and backlog prioritization across parallel projects, balancing new feature development against maintenance of live applications.

Software Engineer | December 2022 – July 2024

- Built the **Portal Logistik API** (Golang + GoFiber + GORM): a modular-monolith service with **338 REST endpoints**, **54 domain models**, and 26 domain modules powering end-to-end procurement — proposals, vendor technical evaluation (EVTEK), winner determination, SLA monitoring, and SAP integration.
- Implemented centralized authentication with **Keycloak** (Authorization Code + OIDC, JWT validation, granular RBAC), replacing separate logins across 50+ applications with a single Single Sign-On identity for 3,000+ internal users.
- Placed **Kong API Gateway** in front of backend services to centralize authentication, rate limiting, routing, and request observability — moving cross-cutting concerns out of application code.
- Wrote **audit logging** middleware capturing every API request and caller identity to a datastore, providing an audit trail for incident investigation and internal compliance.
- Built operational automation: scheduled cron reminders for tasks breaching SLA thresholds, plus Excel/PDF report exports — eliminating manual reconciliation work.
- Developed supporting IAM microservices in Go (encryption, email, queue, data synchronization) so security and notification concerns could scale independently.

SELECTED PROJECTS

Enterprise Single Sign-On Platform — PUSRI

Role: Core contributor (design & implementation)

Stack: Golang 1.24, Keycloak (gocloak v13 Admin API), OpenID Connect, OAuth 2.0, JWT, WebAuthn/Passkeys, GORM, SQL Server, Docker

Scale: 50+ integrated applications · 3,000+ active users · 46 endpoints

- Designed Keycloak realms, clients, and role mappings as the single source of identity for all internal applications, removing per-application credential management.
- Standardized the Authorization Code flow and JWT validation (signature, audience, expiry) at the resource-server layer, with token refresh, session validation (authme / check-auth), and single logout.
- Added **Passkeys/WebAuthn** and **WhatsApp OTP** as alternative authentication factors, extending secure login coverage to non-desktop users.
- Extended the platform to **mobile IAM** and an IAM web portal, so company mobile apps share the same identity and authorization policies as web applications.
- Mapped RBAC onto the organizational structure (work units & job positions) so authorization follows the company hierarchy without per-application manual configuration.

Portal Logistik API — Procurement & Vendor Technical Evaluation (EVTEK SILOGIS)

Role: Backend Engineer (core contributor)

Stack: Golang, GoFiber, GORM, Redis, Kong API Gateway, Docker, GitLab CI/CD, SonarQube, Nexus Registry, Swagger

Scale: 338 REST endpoints · 54 domain models · 26 domain modules

- Built the **EVTEK** vendor technical-evaluation module end to end: submission, role-based evaluator assignment, technical scoring, and winner determination — digitizing a previously document-driven process.
- Made the employee and work-unit master-data services the single source of evaluator data via HTTP service clients, eliminating duplicated personnel records across systems.
- Integrated **SAP** for material and procurement data synchronization, and built SLA monitoring with automated scheduled reminders.
- Authored the API documentation (OpenAPI/Swagger plus per-module Markdown) that serves as the official integration contract for frontend teams.

SIIRU — Routine Investment Monitoring System

Role: Lead Engineer / Backend Architect

Stack: NestJS 11, TypeScript, Prisma 6, PostgreSQL, Redis, S3/MinIO, ExcelJS, pdfmake, Jest, Docker Compose

Scale: 68 REST endpoints · 12 domain modules · 15 data models

- Designed the data schema and end-to-end routine-investment workflow: **proposal → realization → disbursement → budget relocation**, with dedicated history tables capturing an audit trail of every value change.
- Applied **Redis caching** to dashboard and aggregate master-data endpoints to reduce repeated query load on PostgreSQL.
- Built **Excel and PDF** report exports and file uploads to S3/MinIO object storage using **presigned URLs**, keeping large files off the application server.

- Set up layered DTO validation (class-validator + Joi), health-check endpoints, Prisma migrations, and seeding scripts — making deployments safe and repeatable.
- Wrote **Jest unit + e2e (Supertest)** tests and published interactive OpenAPI documentation at `/api/docs` covering all 68 endpoints.

SIHPP — Cost of Goods Manufactured System

Role: Lead Engineer (backend architecture & implementation)

Stack: Golang 1.23, Fiber v2, GORM, SQL Server, JWT, Keycloak SSO, bcrypt, Swagger UI, Docker

Scale: 17 handler modules · 15 data models · 3 product lines

- Built the production-cost calculation backend for three product lines (**Urea, Ammonia, NPK**), covering projected vs. actual production and all cost components: fixed costs, raw-material prices, utilities, planning ratios, and business & financial costs.
- Applied a layered **handler-service-repository** architecture with a consistent JSON response envelope, so adding a new product module requires no changes to existing modules.
- Integrated company SSO and user-role RBAC across all modules.
- Automated schema migration and master-data seeding at startup, removing manual setup steps for every new deployment environment.
- Published interactive OpenAPI documentation (Swagger UI with *Try it out*) at `/docs/` read as the API contract for frontend teams.

Ismail Laundry — Laundry Management Application (Personal Product, Live in Production)

Role: Solo Full-Stack Developer & Product Owner (design → build → deploy → operate)

Stack: Next.js 16, React 19, TypeScript, Prisma 6, PostgreSQL, Tailwind CSS 4, Web Push (PWA), Leaflet, Cloudflare R2, Vercel

Scale: 80+ active customers · 950+ transactions processed · 68 API endpoints · 48 pages · 21 data models

- Independently built and operate an end-to-end laundry management system: orders & invoicing, **installment payments and settlement**, per-service pricing, loyalty and promo programs, pickup requests with chat, bookkeeping/expenses, and GPS-based staff attendance.
- Designed the installment payment flow with a reconcilable balance cache, supported by **backfill** and **repair** scripts to correct historical data without downtime.
- Built data-migration tooling with `--dry / --diff / --sync / --apply` modes, so production migrations are always inspectable before execution.
- Conducted an independent **security review** of the entire application, documenting **17 ranked findings** (Critical → Low) — including unauthenticated admin endpoints, IDOR, arbitrary file read, mass assignment, and hardcoded fallback secrets — then remediated them against a sprint-based roadmap. (*Case study available on request.*)
- Wrote 9 per-domain **smoke-test** scripts (auth, orders, installments, finance, loyalty, promo, pricing, attendance, pickup) as a pre-release regression safety net.

EDUCATION

Universitas Muhammadiyah Palembang — Palembang, Indonesia

Bachelor of Information Technology (S1) | Graduated 2022

CERTIFICATIONS

- **AI Training for Product & Service Prototyping** — Skyletica Labs, May 2026
- **React JS — Web Front End Development** — SanberCode, August 2023
- **Golang Backend Development** — SanberCode, December 2022 (No. 35896/362/SNBR/BOOTCAMP/XII/2022)
- **React Native Expo — Multiplatform Mobile App Development** — Udemy, September 2022

ADDITIONAL INFORMATION

- **Methodology:** Agile/Scrum — sprint planning, estimation, daily standups, retrospectives; code review and technical mentoring.
- **Breadth:** requirements analysis, architecture & data-schema design, implementation, security hardening, deployment, production maintenance.
- **Languages:** Indonesian (native), English (technical reading & writing). **Work preference:** onsite, hybrid, or remote.